

Investment Final Review

Finance 647

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Outline

1. Risk, Return, Sharpe Ratio, and Trading [Notes 3&4]
2. Portfolio Allocation [Notes 5]
3. Mean Variance Analysis and Estimation [Notes 6&7]

Fundamentals

4. CAPM [Notes 9&10]
5. Anomalies [Notes 13]
6. Active Portfolio Management [Notes 14]
7. Factor Models [Notes 15]

Tools

8. Mutual Funds and Hedge Funds [Notes 17]
9. Socially Responsible Investing (ESG) [Notes 18]
10. Household Finance (Life Cycle Investing) [Notes 19]

Applications

Module 1: Risk, Return, Sharpe Ratio, and Trading

- Two risky assets:

- Portfolio return:

$$r_p = \omega_A r_A + \omega_B r_B$$

- Portfolio variance:

$$\sigma_p^2 = \omega_A^2 \sigma_A^2 + \omega_B^2 \sigma_B^2 + 2\omega_A \omega_B \sigma_A \sigma_B \rho_{AB}$$

- One risky asset and one risk-free asset:

- Portfolio return:

$$r_p = \omega_s r_s + (1 - \omega_s) r_f$$

- Portfolio variance:

$$\sigma_p^2 = \omega_s^2 \sigma_s^2$$

- Date frequencies:

- E.g., monthly return (std dev) to annual return (std dev)

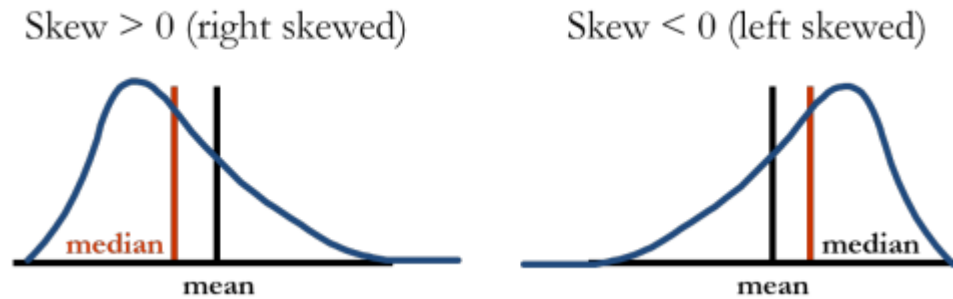
- Sharpe ratio

$$SR_p = \frac{r_p - r_f}{\sigma_p}$$

- What is the Sharpe ratio on US equities in the long run?

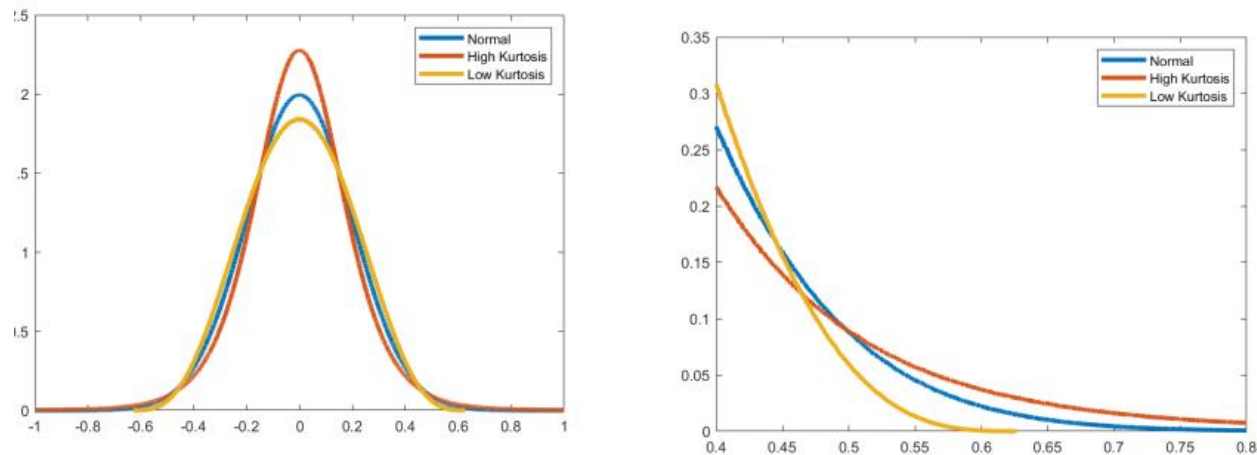
Module 1: Risk, Return, Sharpe Ratio, and Trading

- Skewness:



- Skewness measures the asymmetry of the distribution.

- Kurtosis:



- Kurtosis measures the likelihood of extreme events (thickness of the tails)

Module 1: Risk, Return, Sharpe Ratio, and Trading

- Trading:
 - Bid price, ask price
 - Market order, limit order
 - Order book

Order Book

Top of Book

Bid		Ask	
Price	Size	Price	Size
103.77	200	103.78	700
103.76	600	103.79	2,000
103.75	1,200	103.80	2,100
103.74	2,300	103.81	2,100
103.73	2,200	103.82	1,800

Module 2: Portfolio Allocation

- One risky asset and one risk-free asset.
- Mean-variance optimization:

$$\max r_p - \frac{A}{2} \sigma_p^2 = \max_{\omega_s} \omega_s r_s + (1 - \omega_s) r_f - \frac{A}{2} \omega_s^2 \sigma_s^2$$

- Optimal weight invested in the risky asset:

$$\omega_s^* = \frac{r_s - r_f}{A \sigma_s^2}$$

- Certainty equivalent return:

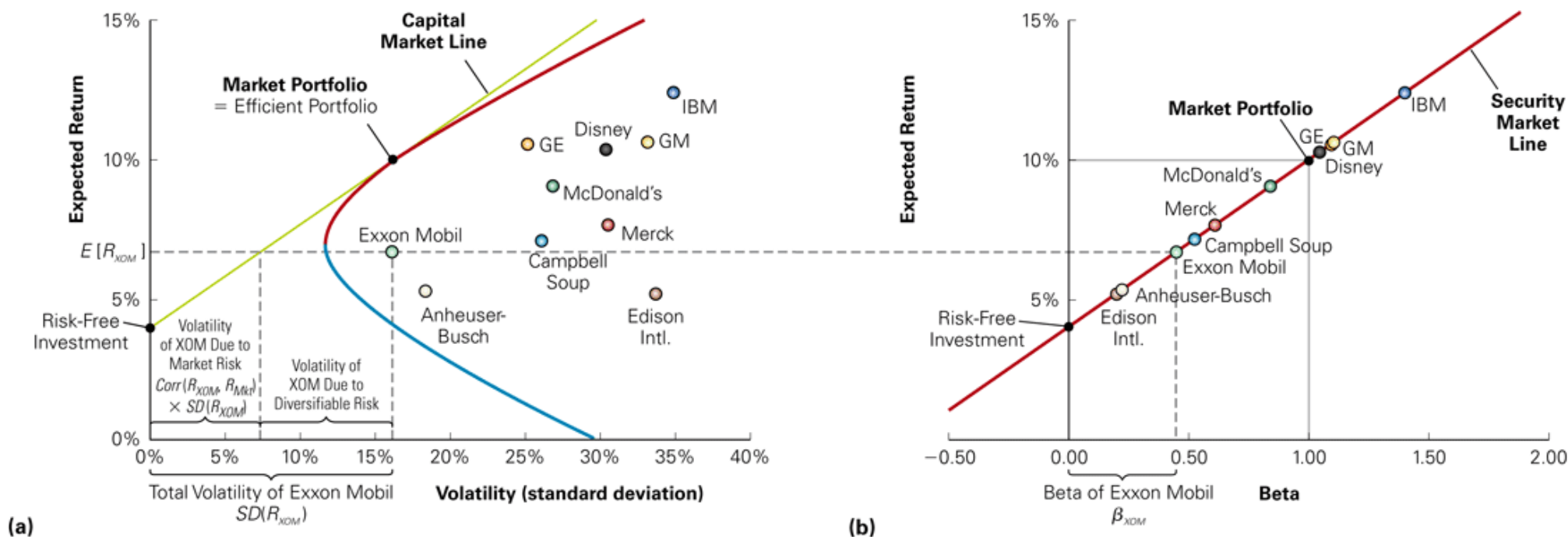
- The maximized utility is:

$$U^* = \omega_s^* r_s + (1 - \omega_s^*) r_f - \frac{A}{2} \omega_s^{*2} \sigma_s^2$$

- U^* is the certainty-equivalent return.
- Estimate the risk aversion coefficient, A .

$$U = \omega_s r_s + (1 - \omega_s) r_f - \frac{A}{2} \omega_s^2 \sigma_s^2$$

Module 3: Mean-Variance Analysis



- Minimum variance frontier, Efficient frontier
- CML:

$$r^{CML} = r_f + SR_M \sigma^{CML}$$

- CML vs SML
- Mutual fund theorem: All MV optimizers hold a mix of the market (tangency) portfolio and cash.

Module 3: Mean-Variance Estimation

- Estimation issues:

- Standard error of mean estimates:

$$SE(\bar{r}) = \frac{\sigma}{\sqrt{T}}$$

- T-statistic of mean estimates:

$$t-stat = \frac{\bar{r}}{SE(\bar{r})}$$

- Confidence interval at the 95% significance level

$$CI = [\bar{r} - 1.96 * SE(\bar{r}) , \bar{r} + 1.96 * SE(\bar{r})]$$

- Accurate estimation of means requires a huge amount of data (long time series).
 - Accurate estimation of volatility requires high-frequency data.

- Alternatives to the mean-variance $(n + n + \frac{n(n-1)}{2})$ framework:

- Minimum variance $(n + \frac{n(n-1)}{2})$
 - Risk parity (n)
 - Equal weights (0)
 - Number of parameters to estimate.

Module 4: CAPM

- Fundamental questions in asset pricing:
 - How should we measure risk?
 - What should be the return that investors receive for bearing the risk?
- Standard deviation is a measure of total risk.
- Total risk = systematic (undiversifiable) risk + idiosyncratic (diversifiable) risk
- CAPM:
 - Systematic risk is what investors care about (do not care about idiosyncratic risk)
 - CAPM beta is a measure of systematic risk, called market risk:

$$\beta_i = \frac{Cov(r_i, r_m)}{Var(r_m)}$$

- Investors are rewarded for bearing (only) the market risk:

$$E[r_i] = r_f + \beta_i(E[r_m] - r_f) \text{ (SML)}$$

- Multi-factor models:
 - The systematic risk includes market risk, value risk, size risk, ...
 - Investors are rewarded for bearing these risks.

$$E[r_i] = r_f + \beta_M * MRP + \beta_S * SMB + \beta_V * HML + \dots$$

Module 4: CAPM

- Interpretations of CAPM regressions:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i(R_{m,t} - R_{f,t}) + \epsilon_{i,t}$$

- r-square: 80% of the variation in returns is explained by the variation in the market factor (or is explained by CAPM).
- $\beta_i(R_{m,t} - R_{f,t})$ is the systematic part of asset i's return.
- $\alpha_i + \epsilon_{i,t}$ is the idiosyncratic part of asset i's return.
- If CAPM holds, $\alpha_i = 0$.
 - If $\alpha_i > 0$, this asset has extra positive returns. (Underpriced according to CAPM)
 - If $\alpha_i < 0$, this asset has extra negative returns. (Overpriced according to CAPM)
- This regression is used to estimate the market beta.
- Take the variance of both sides:

$$\sigma_i^2 = \beta_i^2 \cdot \sigma_m^2 + \sigma_{\epsilon_i}^2$$

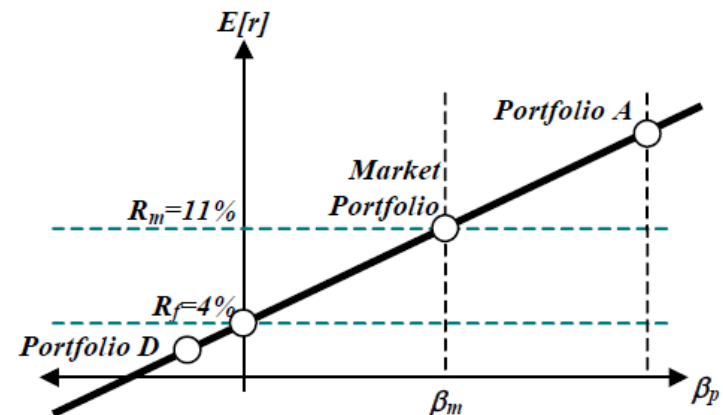
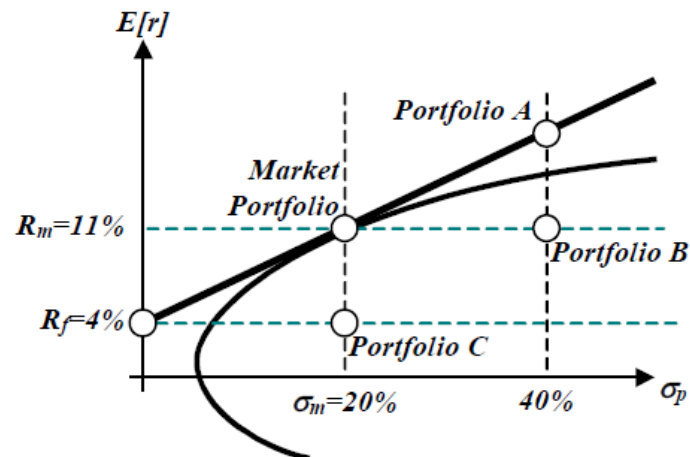
(total risk = systematic risk + idiosyncratic risk)

Module 4: CAPM

- Practice:
 - Class Notes 10, Page 16

Asset	Expected Return	Std. Dev	Corr(R_i, R_m)	Beta
Portfolio 1	0.125	i)	1	1.5
Portfolio 2	0.1625	0.18	0.5	ii)
Portfolio 3	0.07	0.037	iii)	0.5
Market Portfolio	0.10	0.04	iv)	v)
Risk-free	0.05	0	vi)	vii)

- Quiz 2



Module 5: Anomalies

- Motivation: CAPM is not perfectly correct, there are ways to beat the market.
- Beta anomaly:
 - What is it: low (high) beta stocks outperform (underperform) relative to CAPM
 - How to implement: sort by beta (betting against beta)
 - Explanations: leverage constrained story (Frazzini and Pedersenm, 2014), Durations
- Size anomaly:
 - What is it: small stocks outperform big stocks
 - How to implement: sort by size (market capitalization)
 - Explanations: tax-loss selling
- Value anomaly:
 - What is it: value stocks outperform growth stocks
 - How to implement: sort by B/M ratio
 - Explanations: responsiveness to shocks
- Momentum anomaly:
 - What is it: winners outperform losers
 - How to implement: sort by performance in the last 12 months
 - Explanations:

Module 6: Active Portfolio Management

- Sharpe Ratio, Alpha, and Appraisal Ratio
- Sharpe ratio:
 - Measures the return per unit of risk of any stock or portfolio. Maximizing the total portfolio's Sharpe Ratio is the ultimate investor goal under a mean-variance framework.
- Alpha
 - Measures the risk-adjusted return according to some asset pricing model (CAPM or FF 3-factor model).
- Appraisal Ratio

$$AR_i = \frac{\alpha_i}{std(\epsilon_{i,t})}$$

- Measures risk-adjusted return against residual risk.
- It is the Sharpe ratio of the idiosyncratic part of return $(\alpha_i + \epsilon_{i,t})$.
- It is used when considering to add a new investment to an existing portfolio. It determines how much an additional asset improves the original portfolio.

$$SR_{new} = \sqrt{SR_{old}^2 + AR_i^2}$$

Module 7: Factor Models

- CAPM is not good enough, there are other types of systematic risk.

- Carhart 4-factor model:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,M} * MRP_t + \beta_{i,S} * SMB_t + \beta_{i,V} * HML_t + \beta_{i,Mom} * UMD_t + \epsilon_{i,t}$$

- Three factors: MRP, SMB, HML, UMD.
- Interpretations of factor model regressions:
 - E.g., $R_{BR,t} - R_{f,t} = 0.01 + 0.92 * MRP_t + 0.28 * SMB_t + 0.43 * HML_t + \epsilon_{i,t}$.
 - E.g., 0.28 is significant: the loading on SMB is significantly positive, Blackrock is significantly exposed to the size risk, Blackrock mainly invests in small stocks.
 - r-square: 28% of the variation in returns is explained by the variation in the FF 3 factors (or is explained by the FF 3-factor model).
 - Replication: 92% in the market portfolio, 28% in the SMB portfolio, 43% in the HML portfolio, (1-92% = 8%) in the risk-free account.
 - Performance attribution: The FF 3-factor model explains 27% of Blackrock's average returns (the share of returns captured by the factors) .

	Loading	× Factor Mean	= Contribution	Share
alpha			0.16	0.73
Mkt-RF	0.92	0.05	0.05	0.22
SMB	0.28	0.03	0.01	0.04
HML	0.43	0.01	0.00	0.02
		Total	0.2139	1.00

Module 8: Mutual Funds and Hedge Funds

- Empirical facts on mutual funds:
 - Trend of management fees: declining or rising.
 - Performance of actively managed MF: alphas (after fees/before fees), market beta.
 - Managers pick stocks that outperform, but overall negative alphas, why?
 - Small funds tend to outperform large funds, why?
 - MF managers have little performance persistence.
- Empirical facts on hedge funds:
 - Compensations for HF managers (declining): management fees, performance fees.
 - Performance of HF: alphas (after fees/before fees), market beta.
 - Some trading strategies of HF: market neutral strategy, fixed income arbitrage, etc.

Module 9: Social Responsible Investing (ESG)

- Returns on ESG investing, two competing views:
- View 1: ESG investing, by construction, delivers worse risk-adjusted returns.
- View 2: ESG investing delivers better risk-adjusted returns.
 - “Good” stocks have latent (good) cash-flow/risk characteristics that are not priced by the market. (“Good” is good, “bad” is bad)

Module 10: Household Finance (Life Cycle Investing)

- How you should adjust your portfolio over the lifetime:
 - Rebalancing (countercyclical, beneficial),
 - Life cycle theories (NOT required).
- Common investment mistakes:
 - Failure to participate.
 - Failure to diversify.
 - Failure to refinance.